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SUPERSEDING
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MILITARY SPECIFICATION

PACKBOARD, PLYWOOD

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope.- This specification covers one size of plywood packboard.

1.2 Classification.- Plywood packboards shall be of the following types, as specified (see 6.2).

Type I - Packboard, plywood, buckle fasteners on right and left side shoulder straps

Type II - Packboard, plywood, buckle fastener on right side shoulder strap; quick release and quick release billet on left side shoulder strap

2. APPLICABLE DOCUMENTS

* 2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

T-C-571	-	Cords, Cotton; General and Special Purpose, Sash and Venetian Blind
V-T-285	-	Thread, Polyester
QQ-C-390	-	Copper Alloy Castings (Including Cast Bar)
QQ-S-698	-	Steel, Sheet and Strip, Low-Carbon
QQ-S-700	-	Steel Sheet and Strip, Medium and High Carbon
QQ-W-428	-	Wire, Steel, High Carbon, Round, Uncoated for Mechanical Springs (General Purpose)

FSC 8465

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QQ-Z-325 - Zinc Coating, Electrodeposited, Requirements For
TT-C-490 - Cleaning Methods and Pretreatment of Ferrous Surfaces for Organic Coatings
TT-E-529 - Enamel, Alkyd, Semi-Gloss
TT-P-636 - Primer Coating, Alkyd, Wood and Ferrous Metal
CCC-C-419 - Cloth, Duck, Cotton, Unbleached, Plied-Yarns, Army and Numbered
CCC-D-950 - Dyeing and Aftertreating Processes for Cotton Cloths
DDD-L-20 - Label; for Clothing, Equipage, and Tentage, (General Use)
PPP-B-636 - Box, Fiberboard
PPP-F-320 - Fiberboard; Corrugated and Solid, Sheet Stock (Container Grade), and Cut Shapes

MILITARY

MIL-F-495 - Finish, Chemical, Black, for Copper Alloys
MIL-W-530 - Webbing, Textile, Cotton, General Purpose, Natural or in Colors
MIL-B-543 - Buckles, Tongueless and Web Strap
MIL-R-3390 - Rings, Dee
MIL-S-13518 - Wood Preservative: Tetrachlorophenol and Pentachlorophenol, Surface Sealing Compound
MIL-G-16491 - Grommet, Metallic

STANDARDS

FEDERAL

FED-STD-595 - Colors
FED-STD-751 - Stitches, Seams, and Stitchings

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-147 - Palletized and Containerized Unit Loads 40" x 48" Pallets, Skids, Runners, or Pallet-Type Base
MS 51926 - Clips, End, Strap - Type I

DRAWINGS

ARMY NATICK LABORATORIES

3-1-7 - Packboard, Plywood; Type I, Assembly and Details
3-1-8 - Packboard, Plywood; Type I, Details

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3-1-14 - Packboard, Plywood; Type II, Fastener, Quick Release Assembly and Details
3-1-15 - Packboard, Plywood; Type II, Assembly and Details

(Miniature copies of Drawings 3-1-7, 3-1-8, 3-1-14 and 3-1-15, identified as figures 1, 2, 3 and 4 respectively, are attached for information purposes only.)

(Copies of specifications, standards, and drawings required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications.- The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

UNITED STATES DEPARTMENT OF COMMERCE

Commercial Standard CS35 - Hardwood Plywood

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, D.C. 20402.)

UNITED STATES OF AMERICA STANDARD INSTITUTE

USASI Standard B18.1-1965 - Small Solid Rivets

(Copies of USASI Standards may be obtained from the United States of America Standards Institute, 10 East 40th Street, New York, N.Y. 10016.)

NATIONAL CLASSIFICATION BOARD

National Motor Freight Classification

(Application for copies should be addressed to the National Classification Board, 1616 P Street, N.W., Washington, D.C. 20036.)

UNIFORM CLASSIFICATION COMMITTEE

Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification Committee, 202 Union Station, Chicago, Illinois 60606.)

(Technical society and technical association specification and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

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3. REQUIREMENTS

3.1 First article.- Unless otherwise specified (see 6.2), before production is commenced, a sample of the finished specified packboard shall be submitted or made available to the contracting officer or his authorized representative for approval in accordance with 4.2. Approval of the preproduction sample authorizes the commencement of production but does not relieve the supplier of responsibility for compliance with all applicable provisions of this specification. The preproduction sample shall be manufactured in the same facilities to be used for the manufacture of the production packboards.

3.2 Materials.- The materials shall conform to the referenced specifications and requirements specified hereinafter.

3.2.1 Plywood, molded.- The molded or formed plywood construction shall conform to type I of Commercial Standard CS35. The plywood panel shall be 5 or 7 ply. The plywood thickness shall be not less than 0.250 inch and not more than 0.280 inch with face and back veneers conforming to grade 2 or better. The total thickness of even numbered plies shall be not less than 40 percent nor more than 50 percent of the total thickness of the plywood (see 6.3). Direction of grain of odd numbered plies (numbered from face to face) shall be perpendicular to the longitudinal axis of the packboard. Face plies shall be either one piece of veneer or more than one piece provided the glued edges are straight, square, tight and have tapeless spliced joints. Corresponding plies on either side of the core shall be the same thickness and of the same species of wood (general construction). Plywood shall be formed to shape in the curing operations. Formed plywood shall have no ruptured grain at point of curvature. Any of the following species of wood shall be used:

Sweet birch	Yellow birch
Sugar maple	Black maple

3.2.2 Steel sheet or strip.- Steel sheet or strip shall conform to QQ-S-698 or QQ-S-700.

3.2.3 Steel wire.- Steel wire shall conform to type I of QQ-W-428.

* 3.2.4 Brass castings.- Brass castings shall conform to alloy No. A3, type I casting of QQ-C-390.

3.2.5 Duck, cotton.- The cotton duck shall conform to type I, No. 6, hard texture of CCC-C-419. The duck shall be dyed to match the approved shade of Olive Drab 7 (see 6.4). The dyeing and aftertreating process shall conform to type I, class B of CCC-D-950.

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3.2.6 Webbing, cotton.- Cotton webbing shall conform to type IIb, class 4 of MIL-W-530. The webbing shall be dyed to match the approved shade of Olive Drab 7 (see 6.4). The sizes shall be as specified on the drawing.

3.2.7 Thread, polyester.- The thread shall conform to type I, class 1, subclass B, Olive Drab S-1, C.A. 66022 of V-T-285. The dyed thread shall show fastness to chlorine bleaching equal to or better than the standard sample (see 6.4). When a standard sample is not available, the thread shall show good fastness to chlorine bleaching. The letter size shall be as follows:

For all stitching except overcast stitching - Size FF

Overcast stitching - Size E

3.2.8 Cord (cotton, braided) lacing and lashing.- The lacing cord and the lashing cord shall conform to type I, class 3 of T-C-571. The lacing cord shall be size No. 4, and the lashing cord shall be size No. 6.

3.2.9 Primer.- Primer shall conform to TT-P-636.

3.2.10 Sealer.- Sealer shall conform to type II, class 2 of MIL-S-13518.

3.2.11 Enamel.- Enamel shall conform to class A of TT-E-529. The color shall be Olive Drab No. X24087 of FED-STD-595.

3.3 Detail of components.-

3.3.1 Grommets with spur washers.- Grommets with spur washers shall conform to type III, class 3, size No. 0 of MIL-G-16491.

3.3.2 Buckles, tongueless, double bar.- The double bar tongueless buckles shall conform to type II, style 1, class 1, size 1-inch of MIL-B-543.

3.3.3 Buckles, tongueless, single bar.- The single bar tongueless buckles shall conform to type I, style 2, class 1, size 1-inch of MIL-B-543.

3.3.4 Clips, end strap.- The strap end clips shall be fabricated of brass and shall conform to type I, size 1-inch of MS 51926.

* 3.3.5 Ring, dee.- The dee ring shall be brass and shall conform to class 1, symbol D, of MIL-R-3390.

3.3.6 Hooks, lashing and clip, rifle pin.- The lashing hooks and the rifle pin clip shall be fabricated of steel specified in 3.2.2 and shall conform to Drawing 3-1-7.

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3.3.7 Pin, rifle and stiffener, cover.- The rifle pin and cover stiffener shall be fabricated from steel wire specified in 3.2.3. The rifle pin shall conform to Drawing 3-1-7, and the cover stiffener shall conform to Drawing 3-1-8.

3.3.8 Release, quick.- The quick release shall be cast of brass specified in 3.2.4 and shall conform to Drawing 3-1-14.

3.3.9 Rivet, truss head.- Truss head rivets shall be low carbon steel and shall conform to USASI Standard B 18.1 - 1965 and to Drawing 3-1-7.

3.4 Construction.- Construction of the type I plywood packboard shall be in accordance with the requirements of Drawings 3-1-7 and 3-1-8 and as specified hereinafter. The construction of the type II plywood packboard shall be in accordance with requirements of Drawings 3-1-7 and 3-1-8 except that left shoulder strap and left shoulder billet strap shall conform to Drawings 3-1-14 and 3-1-15. Commercial tolerances will be allowed for thickness of metal parts.

3.4.1 Canvas cover.- The canvas cover shall be fabricated with the warp of the duck extending across the packboard in the direction of the lacing tension. When the cover is procured separately, two lacing cords shall be furnished with each cover.

3.4.2 Machining.- All edge corners of the packboard and all edges of openings shall be rounded to a radius of approximately 1/8 inch. The lacing holes shall be rounded on both sides. After machining operations, the packboard shall be sanded to produce a smooth uniform surface, including all edges. Openings on any exposed edge shall be filled with commercial filler and smoothly finished.

3.4.3 Riveting.- All solid rivets shall be set with their heads flush against the wood and peened without fins or rough edges.

3.4.4 Stitching, machine.-

3.4.4.1 Stitch types.- Stitch types shall conform to FED-STD-751 as follows:

For all stitching except overcast stitching	- Type 301, 6 to 8 stitches per inch
For overcast stitching	- Type 304, 14 to 16 stitches per inch

3.4.4.2 Thread breaks.- Thread breaks for all stitching shall be over-stitched not less than one inch at each break.

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3.4.4.3 Stitching ends for stitch type 301.- Stitching at the ends shall be overstitched not less than one inch on stitch type 301 except where ends are turned under in a hem or held down by other stitching. When stitch patterns such as box, box with cross stitch, "W" stitching or straight line tacking are performed automatically, at least three or more tying, overlapping or backstitches shall be used to secure ends of stitching.

3.4.4.4 The thread shall not be lubricated prior to or during sewing.

3.4.5 Setting of end clips.- The webbing shall be inserted the full depth of the clips. The clips shall be securely attached without cutting the webbing. The webbing shall be flat at the point of entrance.

3.4.6 Setting of grommets.- Holes for grommets shall be punched smaller than the outside diameter of the grommet. The grommet shall be forced through the hole and tightly clinched without cutting the material and inserted against the wire stiffener in the hem of the cover.

3.5 Finish.-

3.5.1 Pretreatment of steel components.- Prior to assembly, the lashing hooks, rifle pin clips, and rifle pins shall be pretreated in accordance with type I or II of TT-C-490.

3.5.2 Priming of steel components.- Prior to assembly and following pretreatment conforming to 3.5.1, the lashing hooks, rifle pin clips, and rifle pins shall be given a coat of primer conforming to 3.2.9.

3.5.3 Zinc coating of rivets and stiffeners.- Rivets shall be zinc coated in accordance with type III, class 2 of QQ-Z-325. The stiffeners shall be zinc coated in accordance with type II, class 2 of QQ-Z-325.

3.5.4 Sealer for plywood.- After machining and sanding as specified in 3.4.2, all exposed surfaces shall be cleaned of dust, glue and other foreign matter. Prior to attachment of other components, the plywood shall be immersed in sealer conforming to 3.2.10 for not less than three minutes at a temperature of 70° F (degrees Fahrenheit) minimum to 90° F maximum. When sealer has thoroughly dried, all surfaces shall be scuff sanded.

3.5.5 Enameling.- After attachment of all hardware components and prior to attachment of the webbing and canvas back, all surfaces of the packboard shall be coated with enamel conforming to 3.2.11. The enamel shall cover all wood and all metal parts, and shall be thoroughly dry before assembly of the canvas back.

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3.5.6 Black chemical finish for quick release.- The quick release specified in 3.3.8 shall be given a black chemical finish conforming to MIL-F-495.

3.6 Assembly.- The packboard shall be assembled as shown on applicable drawing. The canvas back shall be tightly laced in position. Lacing shall start at the top of the canvas back and finish with cord in position shown on drawing and with final knot at the bottom end of the canvas back. Lashing cord shall be coiled and tied with twine or tape between the canvas back and the formed plywood. Shoulder straps shall be pulled through packboard openings and buckled as shown on applicable drawing. Spread of plywood packboard flanges and overall width across flanges with assembly and lacing complete, shall comply with dimensions shown on applicable drawing.

3.7 Resistance to applied static loads.- There shall be no fracture or ruptured grain of the plywood when a 600 pound load is applied to the packboard when tested as specified in 4.4.1. The distance between the plywood and canvas back shall be not less than 1-1/4 inches when 100 pounds of weight are distributed over area of the canvas back when tested as specified in 4.4.2. The warp shall be not more than 3/4 inch and the crown shall be not more than 3/8 inch when measured under a compressive load, when tested as specified in 4.4.3.

3.8 Marking for identification.-

3.8.1 Packboard.- The letters U.S. as shown on applicable drawing shall be indented on the face of the packboard before application of finish material. Location and size shall be as shown on applicable drawing. Indenture shall be sufficiently deep to be clearly legible after finish has been applied, but shall not rupture the face of the veneer. The finished face of the packboard shall have the following information stenciled on it with waterproof ink in accordance with the applicable drawing:

(a) Name of supplier.

(b) Contract number and contract date.

3.8.1.1 Ink markings.- The ink markings on the finish face of the packboard shall not smear immediately after removal from the water when rubbed with a finger, be legible, retain their original color, and not crack or peel when subjected to the test specified in 4.4.4.

3.8.2 Canvas cover.- Marking on the canvas cover shall conform to type IV of DDD-L-20. The letters U.S., nomenclature, and identification marks shall conform to class 9 in the size characters and locations shown on applicable drawing. Identification data and fastness of marking shall be as specified for class 5 labels.

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3.9 Workmanship.- Metal components shall be free of burrs, slivers, rough or sharp edges, and shall not be broken, cracked or malformed. Thread tension shall be maintained to prevent loose stitching. All thread ends shall be trimmed. Cloth components shall be clean and free of holes, cuts, tears, or cloth defects such as smashes, weak places, broken or missing yarns. Webbing shall contain no frayed or scalloped edges, and cord components shall be free of cuts or abrasions. Enamel shall level out to a uniform dry film thickness without runs, drops or areas of no film. The coating shall be free from imbedded foreign matter and shall show no evidence of peeling, flaking, or blistering.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

* 4.2 First article inspection.- When a preproduction sample is required, it shall be examined for the defects specified in table V and dimensions specified. The preproduction sample shall be tested as specified in 4.4.1, 4.4.2, 4.4.3, and 4.4.4. Any visual defect, any dimension exceeding specified requirements, or any failure to pass tests shall be cause for rejection of the preproduction sample.

* 4.3 Inspection.- Sampling for inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated hereinafter.

4.3.1 Component and material inspection.- In accordance with 4.1 above, components and materials shall be inspected and tested in accordance with all the requirements of referenced specifications, drawings, and standards unless otherwise excluded, amended, modified or qualified in this specification or applicable purchase document.

4.3.2 In-process inspection.-

4.3.2.1 Process examination.- Examination shall be made during the manufacturing process for the requirements specified in table I to establish that no deviation is made from indicated requirements. Whenever nonconformance is noted, correction shall be made to affected items and process.

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TABLE I.- Process examination

Requirement	Requirement paragraph
Plywood formed to shape in curing operation	3.2.1
Grommet holes punched	3.4.6
Pretreatment of steel components	3.5.1
Priming of steel components	3.5.2
Immersion of plywood in sealer	3.5.4
Sanding of plywood after sealing	3.5.4

4.3.2.2 Examination of plywood veneers.- An examination shall be made of individual veneer thickness of core, crossbands and faces to determine if the percent of total thickness of even numbered plies is in compliance with 3.2.1. The sample unit shall be the core, crossbands, and face veneers of one plywood panel prior to bonding. The inspection level shall be S-3 with an acceptable quality level (AQL) of 4.0 expressed in terms of defects per hundred units.

4.3.2.3 Visual examination of intermediate product.-

4.3.2.3.1 Molded plywood.- Examination shall be made of the molded plywood for defects in table II. The sample unit shall be one plywood packboard after machining and sanding but prior to immersion in sealer. The inspection level shall be S-4 with an AQL of 1.5 for major defects and 6.5 for total defects, expressed in terms of defects per hundred units.

TABLE II.- Visual examination of molded plywood

Examine	Defect	Classification Major	Classification Minor
Plywood	Grain direction of odd numbered plies not as specified	X	
	Composite face veneer not joined (when applicable)	X	

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TABLE II.- Visual examination of molded plywood (cont'd)

Examine	Defect	Classification Major	Classification Minor
Plywood (cont'd)	Joints (when applicable) not straight, square, tight, and tapeless spliced		X
	Ruptured grain	X	
	Not sanded smooth and uniform		X
	Opening in edge of plywood not filled and smoothly finished		X
	Evidence of delamination	X	
	Splintered, chipped, split or cracked	X	

4.3.2.3.2 Visual examination of cover stiffener.- Examination shall be made of the cover stiffener for defects in table III. The sample unit shall be one completely fabricated cover stiffener. The inspection level shall be S-3 with an AQL of 4.0 expressed in terms of defects per hundred units.

TABLE III.- Visual examination of cover stiffener

Examine	Defect
Finish	Not as specified
Workmanship	Burrs, slivers, rough or sharp edges
	Broken, cracked, or malformed

4.3.2.4 Dimensional examination of cover stiffener.- An examination shall be made of cover stiffeners for compliance with dimensions specified. The sample unit shall be one completely fabricated cover stiffener. The inspection level shall be S-3 with an AQL of 4.0 expressed in terms of defects per hundred units.

4.3.2.5 Testing of intermediate product.- The partially fabricated plywood frame shall be tested in accordance with table IV to determine compliance with requirements of Commercial Standard CS35, except that the sample unit shall be one plywood frame after machining but prior to immersion in sealer. The thirty (30) test specimens shall be taken from the areas between the openings of the uncurved portion of the frame.

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TABLE IV.- Testing of the intermediate product

Characteristic	Requirement para.	Test method	Requirement applicable to: Ind. specimen	Requirement applicable to: Panel average	No. of determinations per sample unit	Results reported as	Inspection level	AQL
Dry, shear test	3.2.1	CS35	X	-	15	Percent minimum wood failure	S-1	6.5
			-	X		Percent average wood failure	S-1	-
Cyclic boil test	3.2.1	CS35	X	-	15	Pass or fail	S-1	6.5
			X	-		Percent minimum wood failure	S-1	6.5
			-	X		Percent average wood failure	S-1	-

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4.3.3 End item inspection.- The lot shall be all of the packboards of one type offered for inspection at one time. The sample unit shall be one completely fabricated and assembled packboard.

4.3.3.1 Visual examination.- Examination shall be made of end item for defects in table V. The inspection level shall be level II with an AQL of 2.5 for major defects and 6.5 for total defects, expressed in terms of defects per hundred units.

TABLE V.- Classification of defects

Examine	Defect	Classification Major	Minor
Finish:			
Plywood and metal components	Enamel or chemical finish, as applicable, missing	X	
	Not color specified	X	
	Not uniform		X
	Area of no film	X	
	Runs or drops		X
	Foreign matter imbedded		X
	Peeling, flaking, or blistering	X	
	Not completely dry		X
Textile components	Not color specified		X
Construction and workmanship	Any component missing	X	
	Any component broken, cracked or malformed	X	
	Any component with burrs, slivers, rough or sharp edges	X	
Plywood	Cross break, ruptured or loosened grain	X	
	Opening, hole, or aperture missing	X	
	Lacing holes not rounded on both sides		X
	Surface not smooth		X
	Edges of plywood and openings not rounded		X

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TABLE V.- Classification of defects (cont'd)

Examine	Defect	Classification Major	Minor
Canvas cover	Not fabricated of one piece	X	
	Warp not in direction of lacing tension	X	
	Hole, cut, or tear	X	
	Any smash, broken or missing yarns, or weak places	X	
Webbing	Frayed or scalloped edges		X
Grommets	Not tightly clinched		X
	Material cut or damaged by assembly of grommet	X	
Buckles and dee ring	Reversed, or will not accommodate webbing	X	
End clips	Webbing not secure or not inserted full length of clips	X	
	Webbing not flat at entrance of clip		X
	Material cut or damaged by assembly of end clip to webbing	X	
Lacing and lashing cords	Cut or abrasion	X	
	Ends not stitched as specified on drawing		X
Lashing hook, rifle pin clips, and rifle pins	Loose or not securely attached	X	
Stitches and stitching	Not hemmed where required	X	
	Not type stitch specified	X	
	Stitching margin irregular		X
	More than eight, or less than six stitches per inch (type 301)		X
	More than sixteen or less than fourteen stitches per inch (type 304)		X
	Thread breaks not overstitched 1 inch minimum		X

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TABLE V.- Classification of defects (cont'd)

Examine	Defect	Classification Major	Classification Minor
Stitches and stitching (cont'd)	Type 301 stitching not overstitched 1 inch minimum at ends (except where ends are turned under in a hem or held down by other stitching)		X
	Secured with less than 3 tying, overlapping or backstitches when automatic stitching is performed		X
	Thread ends not trimmed		X
Rivets and riveting	Rivet head not flush against wood		X
	Not peened or is loose	X	
Assembly	Canvas back not tightly laced	X	
	Lacing does not start at top of canvas back		X
	Lacing cord not knotted or secured as specified		X
	Lashing cord not secured in specified location	X	
	Lashing cord not secured by twine or tape		X
	Shoulder straps not pulled through packboard openings and buckled as specified		X
	Two lacing cords not furnished with each cover (when cover is procured separately)	X	
Quick release (when applicable)	On wrong side of strap	X	
	Reversed	X	
	Does not function properly	X	
Marking	Missing, incomplete, misspelled, illegible, of improper type, size or location		X
	The letters "U.S." stamped after application of enamel		X
	Veneer face ruptured by stamping	X	

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4.3.3.2 Dimensional examination.- Examination shall be made of the packboards for dimensions specified. Any noncompliance therewith shall constitute a defect. The inspection level shall be S-3 with an AQL of 4.0 expressed in terms of defects per hundred units.

4.3.3.3 Testing of end item.- Testing of the completely fabricated packboards shall be in accordance with table VI for the characteristics shown therein. The sample unit shall be one completely fabricated packboard with cover. The inspection level shall be S-1 with an AQL of 6.5 expressed in terms of defects per hundred units.

TABLE VI.- Testing of end item

Characteristic	Requirement paragraph	Test method	Number determinations per sample unit	Results reported as
Packboard compression	3.7	4.4.1	1	Pass or fail
Lacing tension	3.7	4.4.2	1	Nearest 1/32 inch
Warp and crown	3.7	4.4.3	2 (one for warp one for crown)	Nearest 1/32 inch
Ink marking	3.8.1.1	4.4.4	1	Pass or fail

4.3.4 Examination of preparation for delivery requirements.- An examination shall be made to determine that packaging, packing, and marking requirements of section 5 of this specification are complied with. Defects shall be scored in accordance with table VII. The sample unit shall be one shipping container fully prepared for delivery with the exception that it need not be sealed. The lot shall be the number of shipping containers offered for delivery at one time. The inspection level shall be S-2 and the AQL shall be 2.5 expressed in terms of defects per hundred units.

TABLE VII.- Examination of preparation for delivery

Examine	Defect
Marking	Omitted; incorrect; illegible; of improper size, location, sequence or method of application

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TABLE VII.- Examination of preparation for delivery (cont'd)

Examine	Defect
Materials	Any component missing or damaged
Workmanship	Inadequate application of components such as incomplete closure of container flaps, loose strapping or tape banding, and inadequate stapling Bulged or distorted container
Contents	Number per package is more or less than required

4.3.4.1 Examination of shipping containers.- When shipping containers are required to be in accordance with PPP-B-636, examination for defects in closure, waterproofing and reinforcing shall be made in accordance with the appendix of that specification.

* 4.3.5 Examination of palletization.- An examination shall be made to determine that palletization is in compliance with section 5 of the specification. Defects shall be scored as specified in table VIII. The sample unit for this examination shall be one palletized unit load ready for shipment. The lot shall be the number of palletized unit loads submitted for inspection at one time. The inspection level shall be S-1 with an AQL of 6.5 expressed in terms of defects per hundred units.

TABLE VIII.- Examination of palletization

Examination	Defect
Finished dimensions	Length, width, or height exceeds specified maximum requirement
Palletization	Not as specified Pallet pattern not as specified Interlocking of loads not as specified Load not bonded with required straps as specified
Weight	Exceeds maximum load limit
Marking	Omitted; incorrect; illegible; of improper size, location sequence or method of application

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4.4 Tests.-

4.4.1 Packboard compression test.- The complete assembled packboard shall be placed in a flat position, plywood upward, on a level surface. A load of 600 pounds shall be applied at the longitudinal center of the packboard between the first and second or second and third openings from the bottom of the packboard. The load shall be distributed through a 3-1/2 inch by 4 inch loading block with the 4 inch dimension parallel to the longest direction of packboard. The bottom of the block shall be slightly rounded (.010 inch to .015 inch), and shall have the two outside edges of the block rounded to a radius of 1/8 inch. The load shall be applied for two minutes. The load shall then be removed and the packboard inspected to determine compliance with 3.7.

4.4.2 Measurement of lacing tension.- The finished packboard shall be placed on a flat surface, plywood down. The shoulder straps and the coiled lashing cord shall be removed. A box or board 4 inches wide and not less than twenty inches long shall be placed on the canvas back, longitudinal to the packboard and centered on the canvas. Weights shall be uniformly distributed in the box or on the board until the total load is 100 pounds. At the option of the supplier a 100 pound bar, at least 20 inches long and 4 inches wide with a flat lower surface and a uniform and symmetrical cross section, may be substituted for the box or board with separate weights. The packboard shall then be inspected to determine compliance with 3.7.

4.4.3 Measurement of warp and crown.- The finished packboard shall be laid on a flat surface, plywood down. The shoulder straps and the coiled lashing cord shall be removed. If there is evidence or warp or twist, the hooks on one side of the packboard shall be held firmly against the surface leaving the opposite side free. Measure the maximum opening distance on the free side for compliance with 3.7. The crown of the packboard shall be measured as follows:

(a) Apply tension as specified in 4.4.2 to the canvas, except that the load shall be such as to cause the distance between the plywood and the canvas (under the bar or board) to be not less than 1-1/4 inches nor more than 1-5/8 inches.

(b) Place a 12 inch straight edge across the inside of the packboard on the plywood near the end of the canvas.

(c) At the center of opening between straight edge and plywood, measure the distance between straight edge and plywood to determine conformance with 3.7.

4.4.4 Testing of ink markings.- The finished face of the packboard shall be immersed in water at 78 ± 2° F for 18 hours. After removal from the water the markings shall be examined to determine compliance with 3.8.1.1.

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5. PREPARATION FOR DELIVERY

5.1 Packaging.- Packaging shall be level A or C, as specified (see 6.2).

5.1.1 Level A.-

5.1.1.1 Packboard.- The two rifle pins at the top of the packboard shall be turned downward against the inside of the packboard.

5.1.1.2 Canvas cover.- When procured separately, twenty-five covers, each complete with two lacing cords passed through the grommets and tied with a loose knot, shall be laid flat, one on top of another, with the top cover inverted to form a uniform stack. The free ends of lacing cords shall be positioned on each canvas cover. The stack shall be securely tied with cord or twine to form a bundle. The ties shall be one lengthwise centered on the bundle, and two girthwise approximately 3-1/2 inches from each end.

* 5.1.2 Level C.- Packboards shall be packaged to afford adequate protection against deterioration and physical damage during shipment from the supply source to the first receiving activity. The supplier may use his standard practice when it meets these requirements.

5.2 Packing.- Packing shall be level A, B or C as specified (see 6.2).

5.2.1 Level A.-

* 5.2.1.1 Packboards.- Ten assembled packboards of one type, packaged as specified in 5.1, shall be packed on edge in a snug-fitting shipping container conforming to style RSC, V2s of PPP-B-636. Placement of the packboards shall be as follows: The first packboard shall be placed in the container with canvas side (back) against one end of the container. The second packboard shall be placed against the first packboard with plywood side (face) against the plywood side (face) of the first packboard. The other eight packboards shall be placed in the container alternated end for end with plywood side against canvas side. Closure, waterproofing, and reinforcing with flat strapping or tape banding shall be in accordance with the appendix of the container specification.

* 5.2.1.2 Covers.- Two hundred canvas covers, packaged as specified in 5.1, shall be placed two bundles long, one wide and four deep in a snug-fitting shipping container conforming to style RSC-L, grade V2s of PPP-B-636. The inside of each shipping container shall be fitted with a taped liner conforming to type CF, class weather-resistant, variety DW, grade V15c of PPP-F-320. Closure, waterproofing by means of tape, and reinforcing with flat strapping or tape banding shall be in accordance with the appendix of the container specification.

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5.2.2 Level B.-

5.2.2.1 Packboards.- Ten assembled packboards of one type, packaged as specified in 5.1, shall be packed on edge in a snug-fitting shipping container conforming to style RSC, type CF (variety SW) or SF, class domestic, grade 275 of PPP-B-636. Placement of the packboards shall be as follows: The first packboard shall be placed in the container with the canvas side (back) against one end of the container. The second packboard shall be placed against the first packboard with plywood side (face) against the plywood side (face) of the first packboard. The other eight packboards shall be placed in the container alternated end for end with plywood side against canvas side. Closure shall be in accordance with method II of the appendix of the container specification.

* 5.2.2.2 Covers.- Two hundred canvas covers, packaged as specified in 5.1, shall be placed two bundles long, one wide and four deep in a snug-fitting shipping container conforming to style RSC-L, type CF (variety SW) or SF, class domestic, grade 275 of PPP-B-636. The inside of each container shall be fitted with a one piece taped liner conforming to type CF, class domestic, variety DW, grade 275 of PPP-F-320. Closure shall be in accordance with method II of the appendix of the container specification.

* 5.2.2.2.1 When specified (see 6.2), the shipping container shall be grade V3c or V3s fiberboard box fabricated in accordance with PPP-B-636 and closed in accordance with the appendix of the container specification. The shipping container material may also be grade V4s of PPP-F-320.

5.2.3 Level C.- Packboards, packaged as specified in 5.1, shall be packed in a manner to insure carrier acceptance and safe delivery at destination at the lowest transportation rate for such supplies. Containers shall be in accordance with Uniform Freight Classification Rules or National Motor Freight Classification Rules, as applicable.

5.3 Palletization.- Unless otherwise specified (see 6.2), packboards packed as specified shall be palletized in accordance with load type I of MIL-STD-147. Each prepared load shall be bonded with primary and secondary straps in accordance with bonding means K and L. Pallet patterns shall be in accordance with the appendix of MIL-STD-147. Interlocking of loads shall be effected by reversing the pattern of each course. If the container is of a size which does not conform to any of the pallet patterns specified in MIL-STD-147, the pallet pattern intended to be used shall be submitted to and approved by the contracting officer prior to its use.

5.4 Marking.- In addition to any special marking required by the contract or order, interior packages and shipping containers shall be marked in accordance with MIL-STD-129.

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6. NOTES

6.1 Intended use.- The packboard is intended to permit a soldier to attach and carry a box, package, weapon, or irregular shaped load (see 6.5).

6.2 Ordering data.- Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Type required (see 1.2).
(c) First article.- When a preproduction sample is not required (see 3.1).
(d) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).
(e) When weather-resistant fiberboard containers are required for level B packing (see 5.2.2.2.1).
(f) When palletization is not required (see 5.3).

6.3 Veneer thickness.- Plywood construction recommended is 5-ply, with 0.050 inch thick core and face plies and 0.0625 inch thick crossbands, resulting in a total thickness of odd numbered plies of 0.150 inch and total thickness of even numbered plies of 0.125 inch. If difficulty is encountered in forming the above veneer thickness to the radius required, the plywood should be constructed of seven equal plies 0.039 inch thick, or seven plies with four odd numbered plies 0.03571 inch thick and three even numbered plies 0.039 inch thick. After pressing and sanding the finished plywood shall be within limits specified in 3.2.1.

6.4 Samples of the approved shade of olive drab cotton duck, webbing, and shade S-1 thread may be obtained from the procuring office (see 3.2.5, 3.2.6 and 3.2.7).

6.5 Specification MIL-A-1812 "Shelf, Cargo Support, Packboard, Pressed Steel" covers a steel attachment that may be used to increase the utility of the plywood packboard.

6.6 The margins of this specification have been marked with an asterisk (*) to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and suppliers are cautioned to evaluate the requirements of this document based on the entire content as written irrespective of the marginal notations and relationship to the last previous issue.

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Navy - SA
Air Force - 82

Review activities:

Navy - SA, MC
Air Force - 82, 85

User activity:

Army - MI

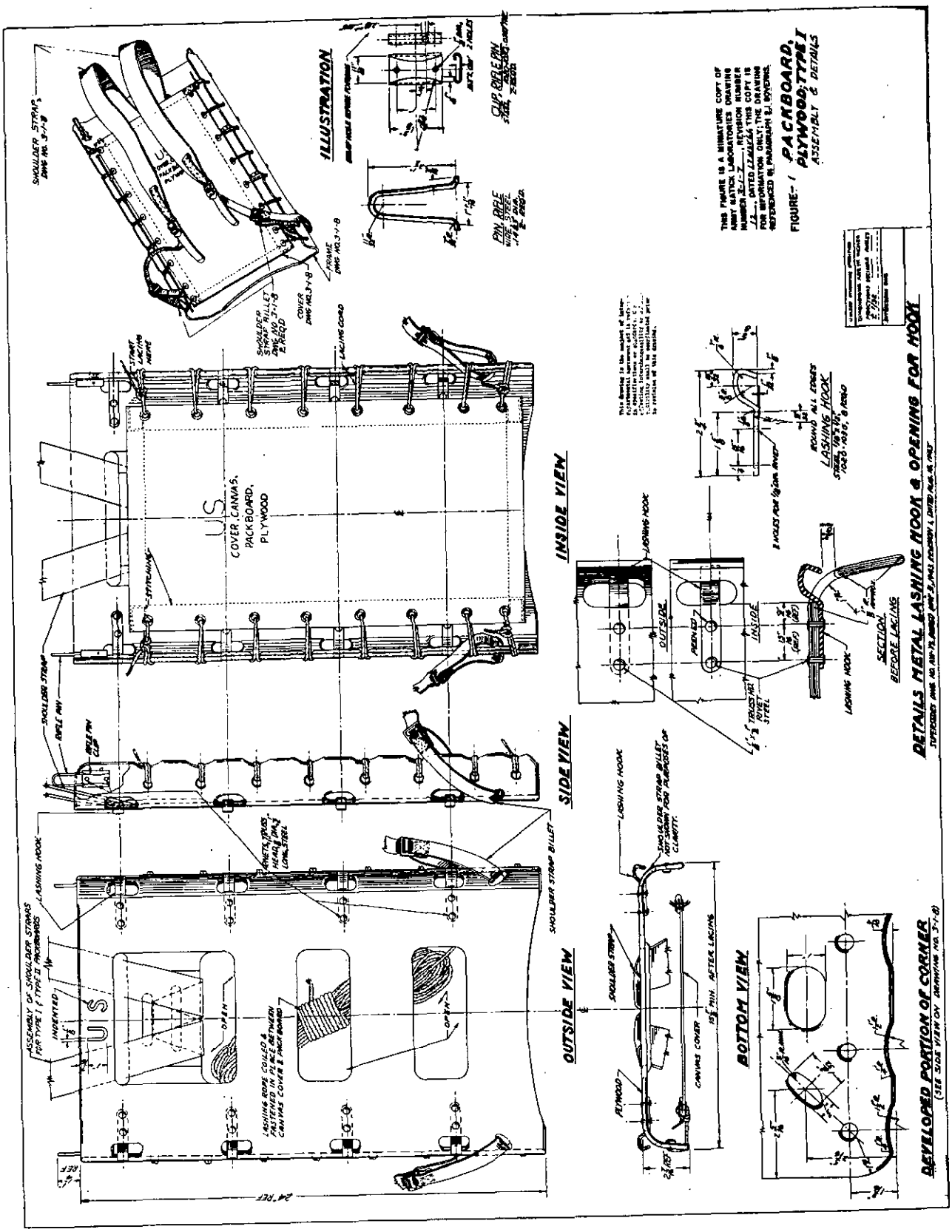
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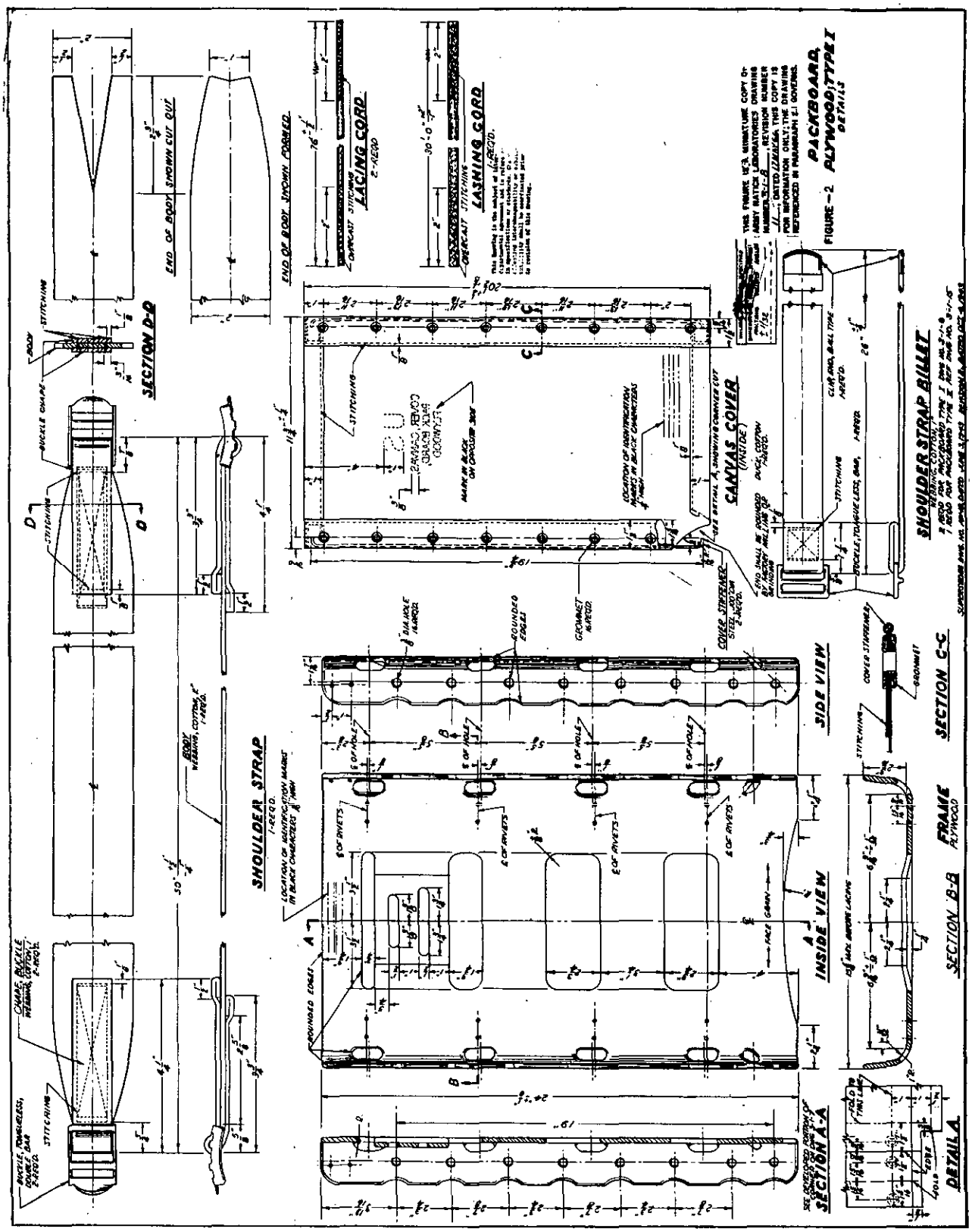
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PACKBOARD, PLYWOOD, TYPE I
FIGURE-2 DETAILS
SHOULDER STRAP BILLET
SHOULDER STRAP
LACING CORD
LASHING CORD
CANVAS COVER
SIDE VIEW
INSIDE VIEW
SECTION C-C
SECTION B-B
FRAME
SECTION A-A
DETAIL A
SECTION D-D
BUCKLE, TONGUELESS, BAR
BUCKLE, TONGUELESS, CHAPE
BUCKLE, TONGUELESS, CHAPE, WELDED
BUCKLE, TONGUELESS, BAR, WELDED

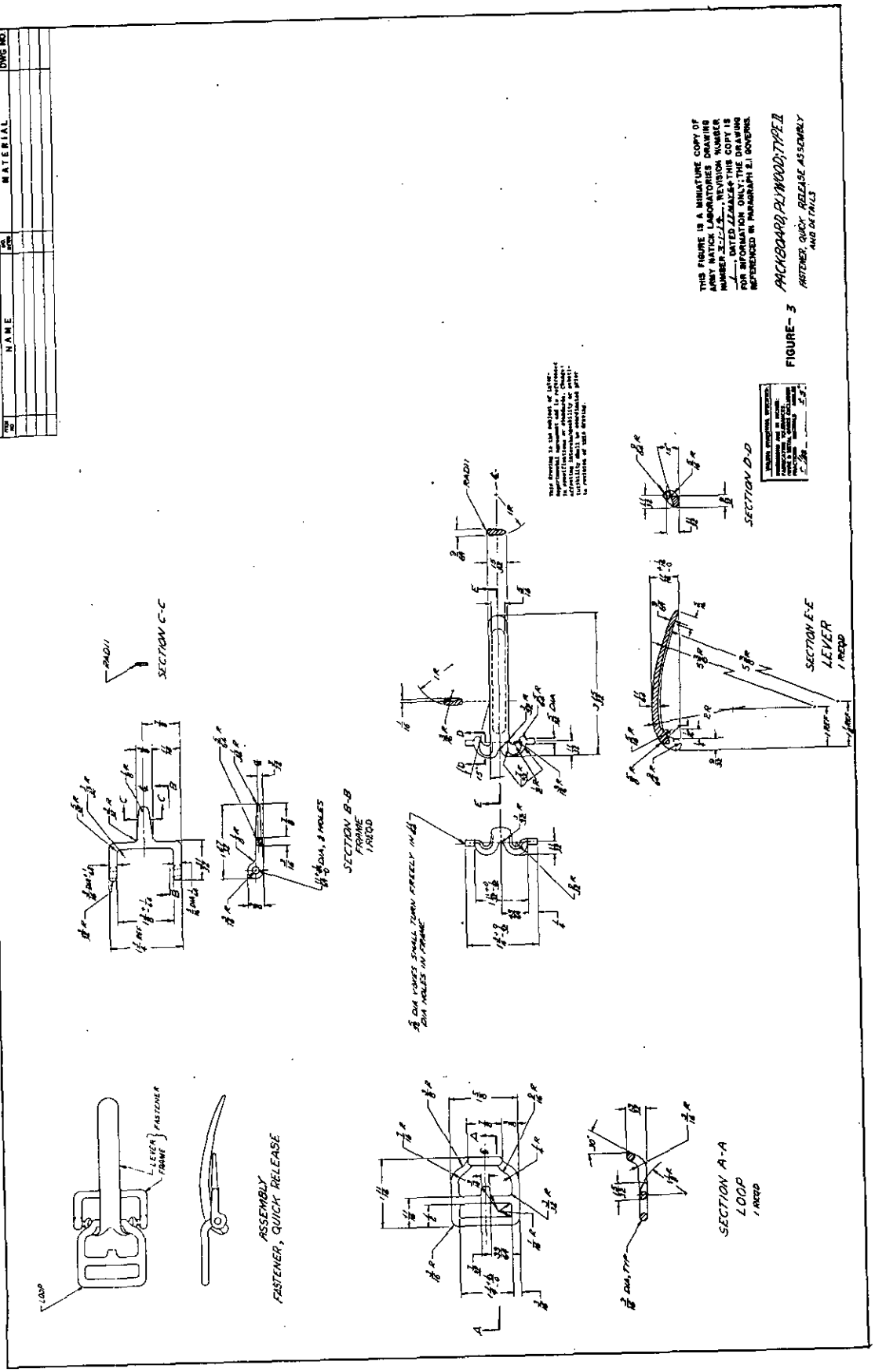
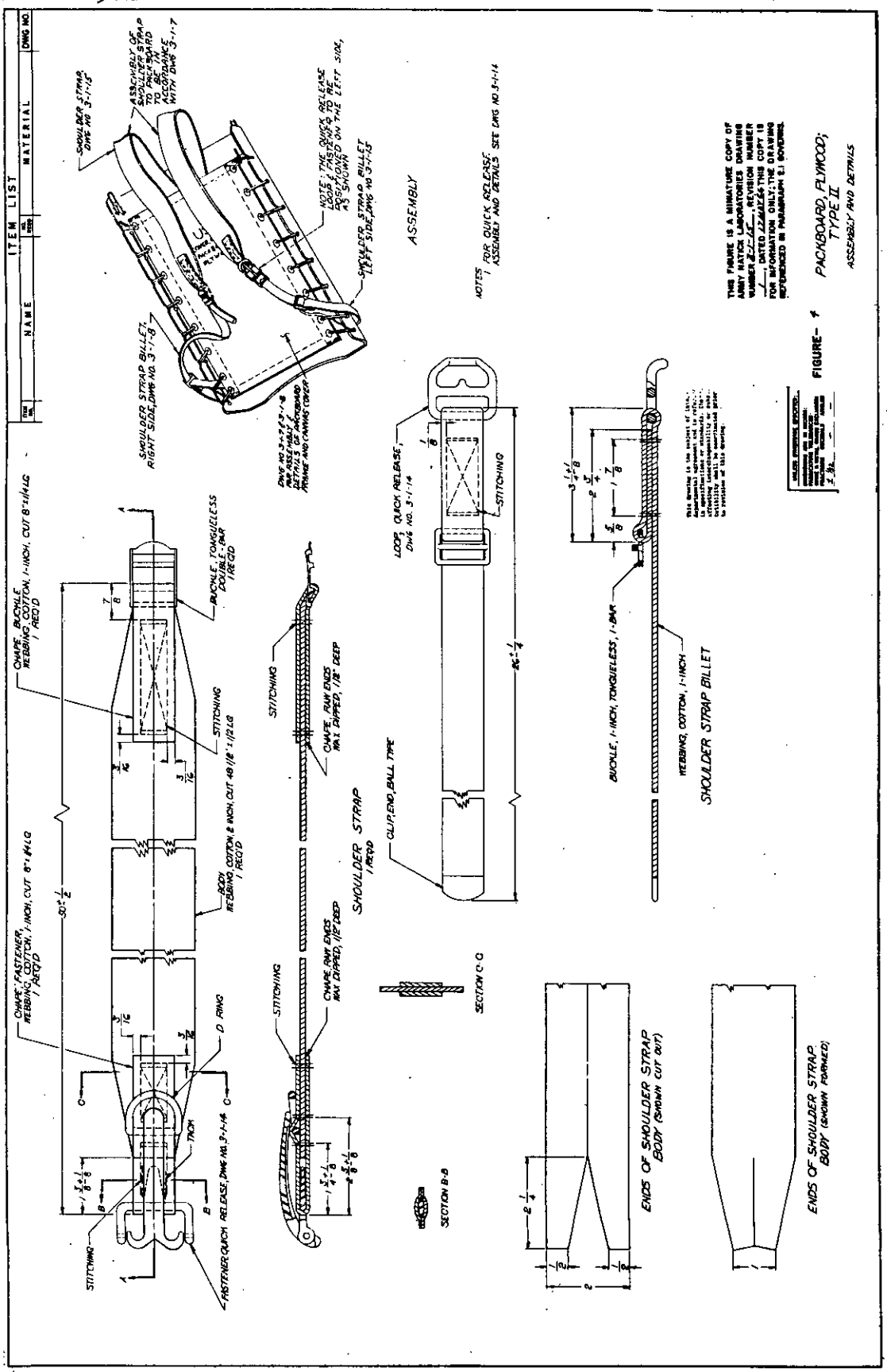


FIGURE 3 PACKBOARD, PLYWOOD, TYPE II
FASTENER, QUICK RELEASE ASSEMBLY AND DETAILS

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION
MIL-P-10941D Packboard, Plywood

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?
   ☐ YES ☐ NO (If "yes", in what way?)

4. REMARKS (Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)

SUBMITTED BY (Printed or typed name and activity - Optional)

DATE

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